

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$x^2 + 7x + 5 = 0$

One solution to the given equation can be written as $x = \frac{-7+\sqrt{k}}{2}$, where k is a constant. What is the value of k ?

Correct Answer: 29

Rationale

The correct answer is **29**. The solutions to a quadratic equation in the form $ax^2 + bx + c = 0$, where a , b , and c are constants, can be calculated using the quadratic formula, which gives two solutions: $x = \frac{-b+\sqrt{b^2-4ac}}{2a}$ and $x = \frac{-b-\sqrt{b^2-4ac}}{2a}$. The given equation is in the form $ax^2 + bx + c = 0$, where $a = 1$, $b = 7$, and $c = 5$. It follows that one solution to the given quadratic equation is $x = \frac{-7+\sqrt{7^2-4(1)(5)}}{2(1)}$, or $x = \frac{-7+\sqrt{49-20}}{2}$, which is equivalent to $x = \frac{-7+\sqrt{29}}{2}$. It's given that one solution to the given equation can be written as $x = \frac{-7+\sqrt{k}}{2}$, where k is a constant. It follows that the value of k is **29**.